CASE STUDY

Cloud Kitchen Performance Dashboard

From 123,141 raw order rows to a three-page Power BI dashboard: cleaning, modelling, metrics and insights

₹38.26M	119,729	₹320	25.55%	2.76%
net revenue	orders delivered	average order value	discount on gross sales	orders cancelled

1. Summary

I analysed one full trading day (22 January 2026) of order data from a multi-brand cloud kitchen network with 51 brands, 409 stores and 115 cities. Orders arrive mainly through Zomato, Swiggy and Eatsure. The raw Excel sheet was cleaned in Python (Pandas), modelled as a star schema, measured with DAX and presented as a three-page Power BI dashboard: an executive overview, a store and risk page, and a brand and operations page.

Three findings stand out.

- **Revenue is concentrated in channels and brands, not in stores.** Zomato and Swiggy bring 84% of net revenue and the top 10 brands bring 83.8%, but the top 10 of 409 stores bring only 6.1%.
- **Discounts are deepest on Eatsure.** Discounts took 25.55% of gross sales (₹13.1M). Eatsure discounts 42.0% against about 22.7% on Zomato and Swiggy, and has the smallest baskets of the three.
- **Cancellations follow the sales channel, not the city.** Among the ten busiest cities the city totals stay between 1.6% and 3.5%, while channels range from 1.6% (Swiggy) to 18.1% (IRCTC). Over half of all cancellations are labelled third-party.

2. Business context and objective

A cloud kitchen network prepares food from delivery-only kitchens and sells several brands from the same store. Management needs to see, in one place, where revenue comes from, how much leaks away through discounts and cancellations, and which stores, channels and brands need attention.

Questions the project answers

- How much did the network earn, and when and where did the revenue come from?
- How much revenue was given away as discounts, and which channels and brands discount most?
- How many orders were cancelled, why, and which channels, hours and stores are most exposed?

3. Data

Item	Detail
Source	Excel workbook, sheet "Raw Data". The other four tabs are Excel practice exercises and were not used.
Size	123,141 rows and 39 columns
Period	Orders of 22 January 2026 (end-of-day date). A few scheduled orders carry later dates, up to 31 January.
Grain	One row per **order_id**, which is one brand's order. The field **order_crn** groups the brand orders of a multi-brand basket.
Main fields	city, store, brand, order source, order type, status, cart amount, discount amount, net amount, coupon text, order date and time, cancellation reason, refund amount
Currency	Assumed to be Indian rupees (₹)

4. Approach

The work followed six steps: profile the data, clean it in Python, model it as a star schema, define the metrics in DAX, build the dashboard, and validate the numbers.

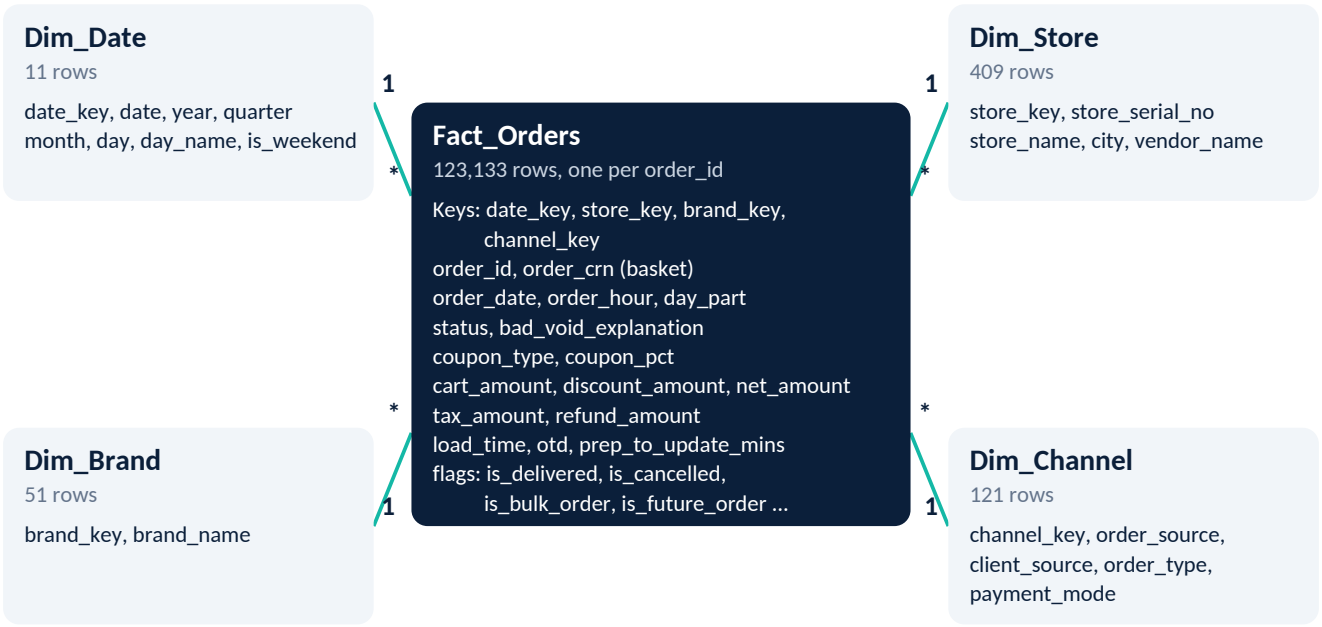
4.1 Data cleaning (Python, Pandas)

Issue found	What I did
8 exact duplicate rows	Removed them, and kept only the latest version of any repeated order_id. 123,133 rows remain.
Two completely empty columns (coupon_code, pos_id)	Dropped them
Typos in headers (paymnet_mode, vender_id)	Renamed to payment_mode and vendor_name
Mixed date formats: order dates are month-first, eod_date is day-first	Parsed each column with its own format into proper date-time values
store_id mixes numbers and codes	Used store_serial_no as the store key, because it is always consistent
coupon_id is free text, for example "40% off + Get items under 89"	Standardised into 7 coupon types plus a discount percentage
Blank refund and delivery charge fields	Filled with 0, because blank means none
Blank delivery-time fields (load_time, otd, promised_time)	Left blank, so averages are not distorted by false zeros
Cancelled, zero-value and bulk orders sit among normal orders	Added flags (is_delivered, is_cancelled, is_zero_value, is_bulk_order, has_discount, is_future_order) instead of deleting rows
No time-of-day field	Derived order_hour and day_part (breakfast, lunch, snacks and evening, dinner, late night)

Two judgement calls: the 802 zero-value orders (mostly the "EatSure Freebies" brand) were flagged, not removed, and bulk orders of up to about ₹1.4 lakh were flagged so that an average excluding them can be shown.

4.2 Data model (star schema)

The flat table was split into one fact table and four dimension tables, exported as CSV files and loaded into Power BI. A star schema keeps filters simple and measures fast, and each dimension can be sliced on its own.



Relationship (many to one)	Direction	Notes
Fact_Orders[store_key] to Dim_Store[store_key]	Single	Store, city, vendor
Fact_Orders[brand_key] to Dim_Brand[brand_key]	Single	Brand
Fact_Orders[channel_key] to Dim_Channel[channel_key]	Single	Junk dimension: order source, client source, order type, payment mode
Fact_Orders[date_key] to Dim_Date[date_key]	Single	Marked as the date table

Filters run from the dimension to the fact table only (single direction), which avoids ambiguous paths. The basket reference order_crn stays in the fact table as a degenerate dimension.

Design choices

- **Grain.** One fact row is one brand order (order_id), because a multi-brand basket holds several brand orders. Basket-level counts use order_crn.
- **Junk dimension.** Dim_Channel combines four low-cardinality fields (order source, client source, order type, payment mode) into one small table.
- **Date table.** Dim_Date has one row per day in the data range and is marked as the date table, so time-intelligence DAX works once more days are loaded.

4.3 Metrics (DAX)

All revenue measures count **delivered orders only**. Revenue is the net amount (cart amount minus discount), because the total_amount field equals net amount plus tax on only about 5% of rows, so its meaning is unclear.

Measure	Definition
Net Revenue	Sum of net_amount for delivered orders
Gross Sales and Discount %	Sum of cart_amount for delivered orders; Discount % = total discount divided by gross sales
Order Count, Basket Count	Distinct order_id and distinct order_crn for delivered orders
AOV	Net Revenue divided by Order Count
Cancellation Rate	Cancelled orders divided by all orders placed
Lost Revenue	Net amount of cancelled orders
Third-Party Cancel Share	Cancelled orders with reason "Third Party Cancellation" divided by all cancelled orders
Median Fulfilment Mins	Median of created-to-last-updated time for delivered, non-scheduled delivery orders
Brands per Basket, Stores Served	Order Count divided by Basket Count; count of all stores, unaffected by slicers

Three of the measures in DAX

```
Net Revenue =
    CALCULATE ( SUM ( Fact_Orders[net_amount] ), Fact_Orders[is_delivered] = 1 )

Cancellation Rate =
    DIVIDE ( [Cancelled Orders], [Orders Placed] )

Third-Party Cancel Share =
    DIVIDE (
        CALCULATE ( [Cancelled Orders],
            Fact_Orders[bad_void_explanation] = "Third Party Cancellation" ),
        [Cancelled Orders] )
```

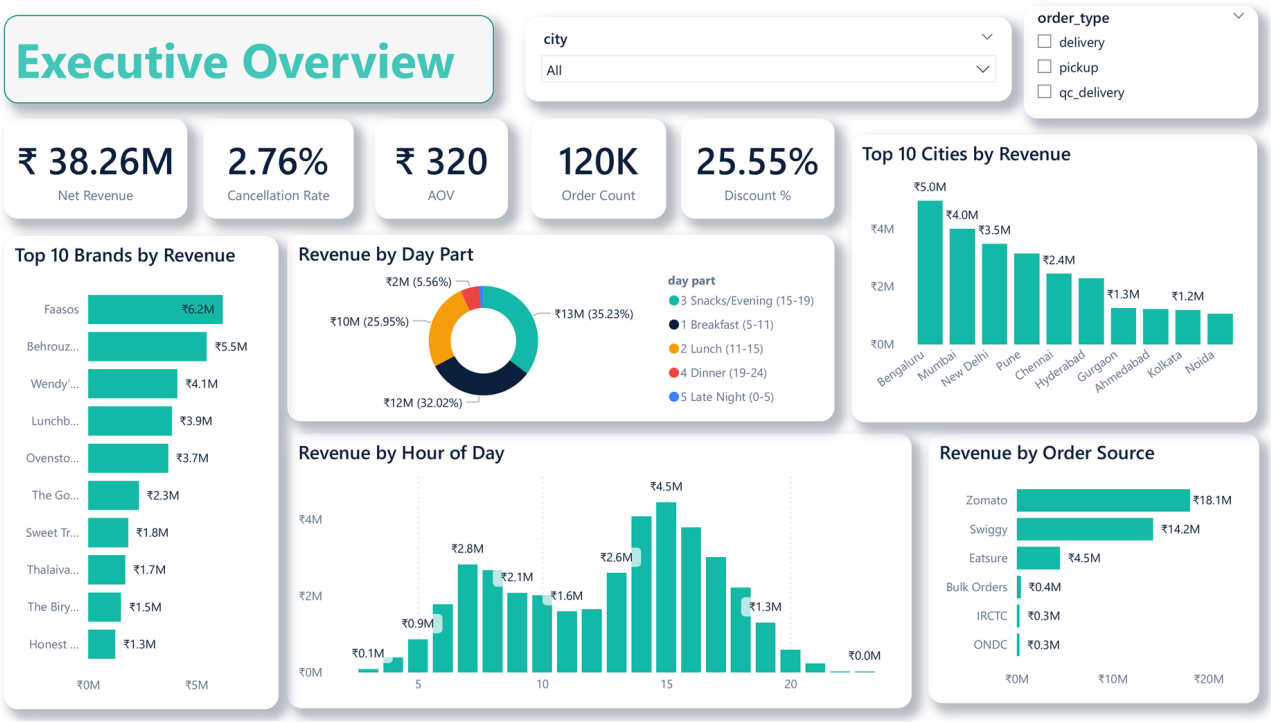
4.4 Validation

- Python checks passed: order_id is unique, no fact row is missing a store, brand, channel or date key, and cart amount minus discount equals net amount on 100% of rows.
- Power BI matched the Python totals: net revenue ₹38,263,719, 119,729 delivered orders, 116,588 baskets and a 2.76% cancellation rate.
- Joining the dimensions back to the fact table keeps the row count unchanged, so no rows are duplicated.

5. Dashboard and insights

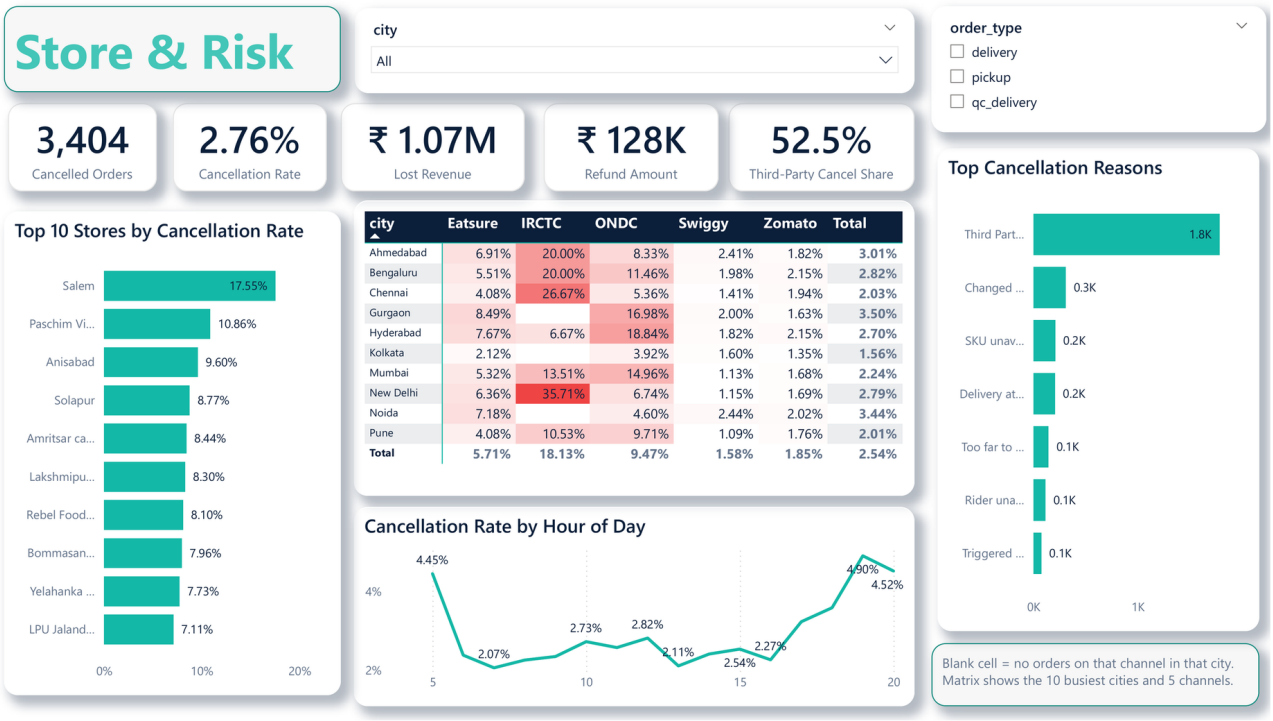
The dashboard has three pages, each answering one question. Every slicer (city, order type, brand) filters all visuals on its page.

5.1 Page 1: Executive Overview



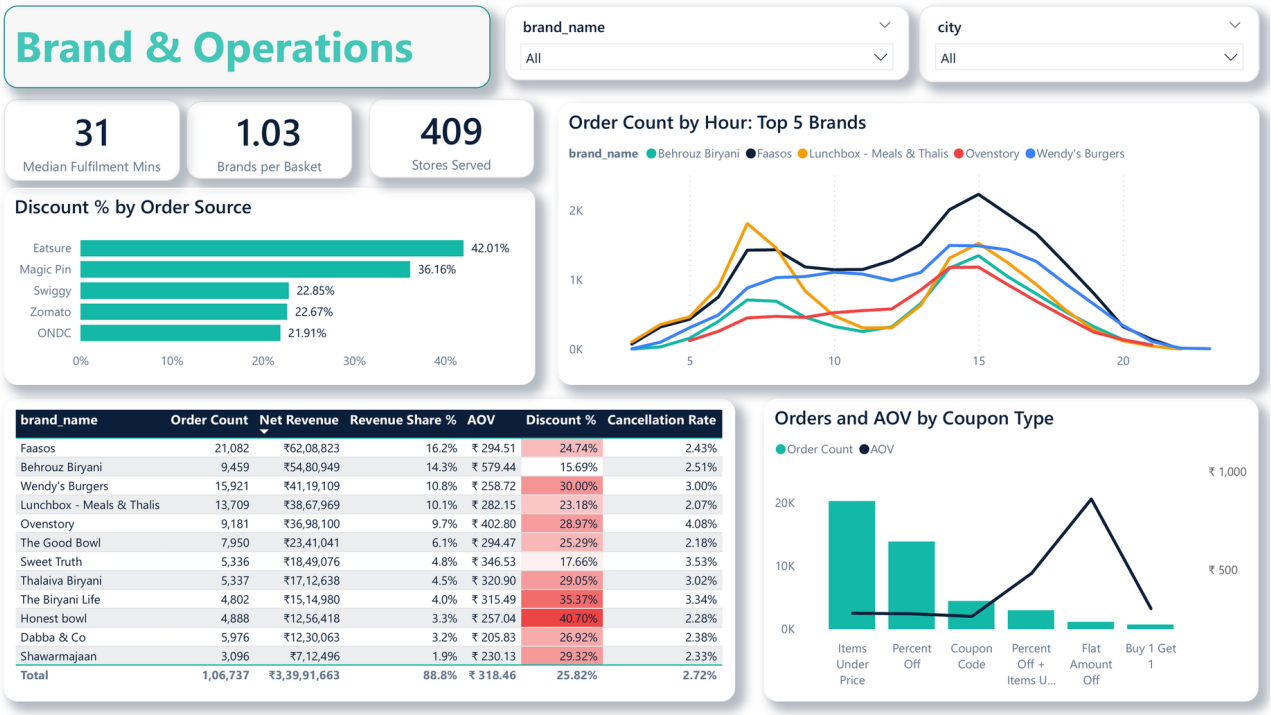
- **₹38.26M net revenue from 119,729 orders.** Gross sales were ₹51.4M, of which ₹13.1M (25.55%) went out as discounts.
- **Two peaks in the day.** Revenue is highest at 3 PM (₹4.5M), with a second peak at 7 to 8 AM (₹2.8M). After 7 PM only 5.6% of revenue arrives, which should be checked against the source system.
- **Where it comes from.** Zomato and Swiggy bring 84% of revenue. Faasos (₹6.2M) and Behrouz Biryani (₹5.5M) lead the brands. Bengaluru (₹5.0M), Mumbai (₹4.0M) and New Delhi (₹3.5M) lead the cities.

5.2 Page 2: Store and Risk



- **3,404 orders were cancelled (2.76%).** That cost ₹1.07M of revenue and ₹128K in refunds. 52.5% of cancellations carry the reason "Third Party Cancellation".
- **The channel matters more than the city.** In the ten busiest cities the totals range only from 1.56% to 3.50%, but by channel IRCTC is at 18.1%, ONDC 9.5%, Eatsure 5.7%, Zomato 1.9% and Swiggy 1.6%. IRCTC and ONDC have few orders there (160 and 1,331), so their rates are indicative. A blank cell means no orders on that channel in that city.
- **Timing and stores.** Cancellations rise at 5 AM (4.45%) and around 7 PM (4.90%). Delivery orders cancel 6.3% of the time, pickup orders 1.9%. Among stores with at least 100 orders, Salem is the outlier at 17.55%, followed by Paschim Vihar (10.86%) and Anisabad (9.60%).

5.3 Page 3: Brand and Operations



- **Discount depth differs widely.** By channel, Eatsure discounts 42.01% and Magic Pin 36.16%, against 22.85% on Swiggy and 22.67% on Zomato. By brand, Honest bowl (40.70%) and The Biryani Life (35.37%) discount most, while Behrouz Biryani discounts 15.69% and has the highest AOV (₹579).
- **Brands peak at different times.** Lunchbox peaks at 7 to 8 AM while Faasos, Behrouz Biryani, Ovenstory and Wendy's Burgers peak at around 3 PM.
- **Coupons go with big baskets.** Orders with flat-amount coupons average about ₹860 against ₹327 without a coupon. This does not prove the coupon raised the basket, because such coupons apply to large carts.
- **Operations.** The median order takes 31 minutes from creation to its last update, and only 1.03 brands are ordered per basket, so multi-brand baskets are rare. The scorecard lists 12 brands that together make 88.8% of revenue; the top 10 make 83.8%.

6. Recommendations

Finding	Recommended action
Eatsure discounts 42% against about 23% on Zomato and Swiggy, with the smallest baskets	Review the depth of Eatsure discounts and test a minimum cart value
52.5% of cancellations are third-party	Share the top cancellation reasons with the Zomato and Swiggy account teams and track them weekly
Salem (17.55%), Paschim Vihar, Anisabad, Solapur and Amritsar cantt cancel far above the 2.76% average	Audit these five stores first
Revenue peaks at 7 to 8 AM and 3 PM, and cancellations rise at 5 AM and around 7 PM	Align staffing and stock with the peaks and check capacity in the evening
Honest bowl and The Biryani Life discount 35 to 41%	Review brand-level discount strategy against margin
IRCTC and ONDC show high rates on small volumes	Keep tracking them on larger volumes before changing anything

7. Limitations and next steps

Limitations

- **One trading day only.** There is no month-on-month, year-on-year or trend analysis. The DAX for those is ready.
- **No customer ID,** so no churn or repeat-order analysis.
- **No item or cost data.** Brand is the lowest level and there is no profit view.
- **Delivery-time fields** (load_time, otd) are unclear and mostly blank, so no on-time claims are made.
- **Evening drop after 7 PM** needs checking against the source system.
- **Small samples** for IRCTC and ONDC. Currency is assumed to be rupees.

Next steps

- Load several months of data and switch on the time-intelligence measures.
- Add a customer ID to measure retention and churn.
- Add item and cost data for menu and margin analysis.
- Automate the Python cleaning and schedule the Power BI refresh.

8. Tools and skills

Area	What I used
Data preparation	Python (Pandas, NumPy) in Google Colab; Excel
Modelling and metrics	Star schema design, Power Query, relationships, DAX measures, a date table
Visualisation	Power BI Desktop: KPI cards, matrix heatmap, ranked bars, column and line charts, slicers, a custom theme
Communication	Three-page dashboard, insights deck in PowerPoint, this case study